

Toward Real-time Generation with MiniMax H3

Li Shunyang Devtech Compute NVIDIA

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Real Time Means Keeping Up

A five-second clip needs a five-second generation budget.

GENERATION TIME

<5 s



GENERATED AUDIO + VIDEO

~5 s

Optimize the complete request, through playable output.

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SM120 Cards

The reference system

RTX PRO 5000

×8

SMs	Memory	Power	Interconnect
110	72 GB GDDR7 ECC	300 W	PCIe Gen5 ×16

No NVLink between GPUs. Communication follows the PCIe and NIC topology.

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Baseline: Latency Breakdown

960 × 544 · 5.167 s generated video

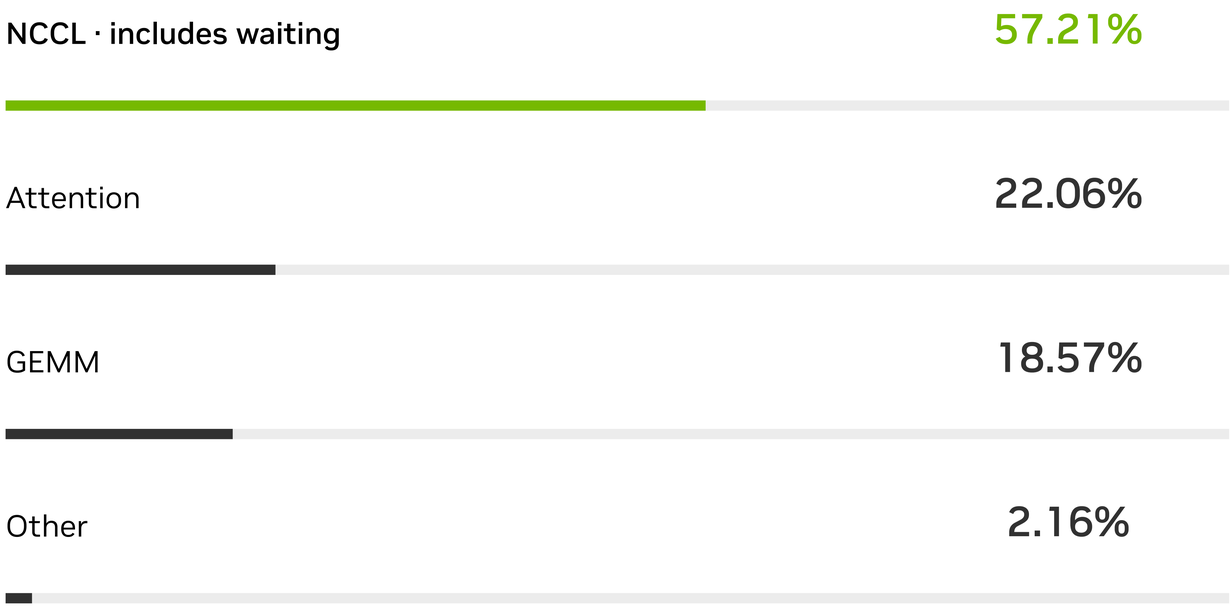
Reference configuration

213.4 s

E2E latency

- 8 × RTX PRO 5000
- 50 steps · BF16 · dense attention
- Ulysses8 · encoder TP8 · VAE tiles8

GPU work by operator



GPU work includes overlap and waiting; these shares are not E2E wall-time components.

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Optimization - Distillation

Reduce how often the model runs.

Base H3

50 model evaluations



Distilled student

8 model evaluations



84%

fewer evaluations

Fewer evaluations reduce compute and communication together.

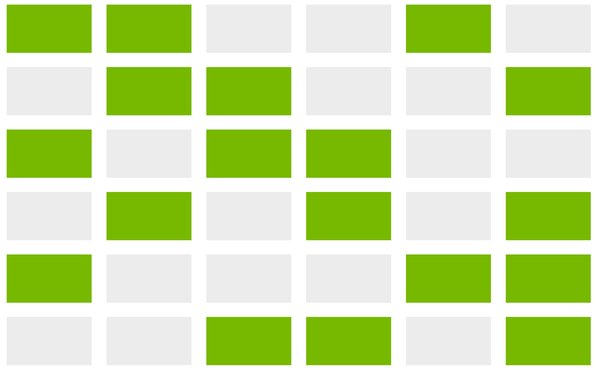
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Optimization - Video Sparse Attention + NVFP4

Do less attention work. Use lower precision where quality allows.

Video Sparse Attention

Attend to selected regions



Content-dependent selection · schematic

NVFP4

Lower precision with QAD

BF16 → NVFP4

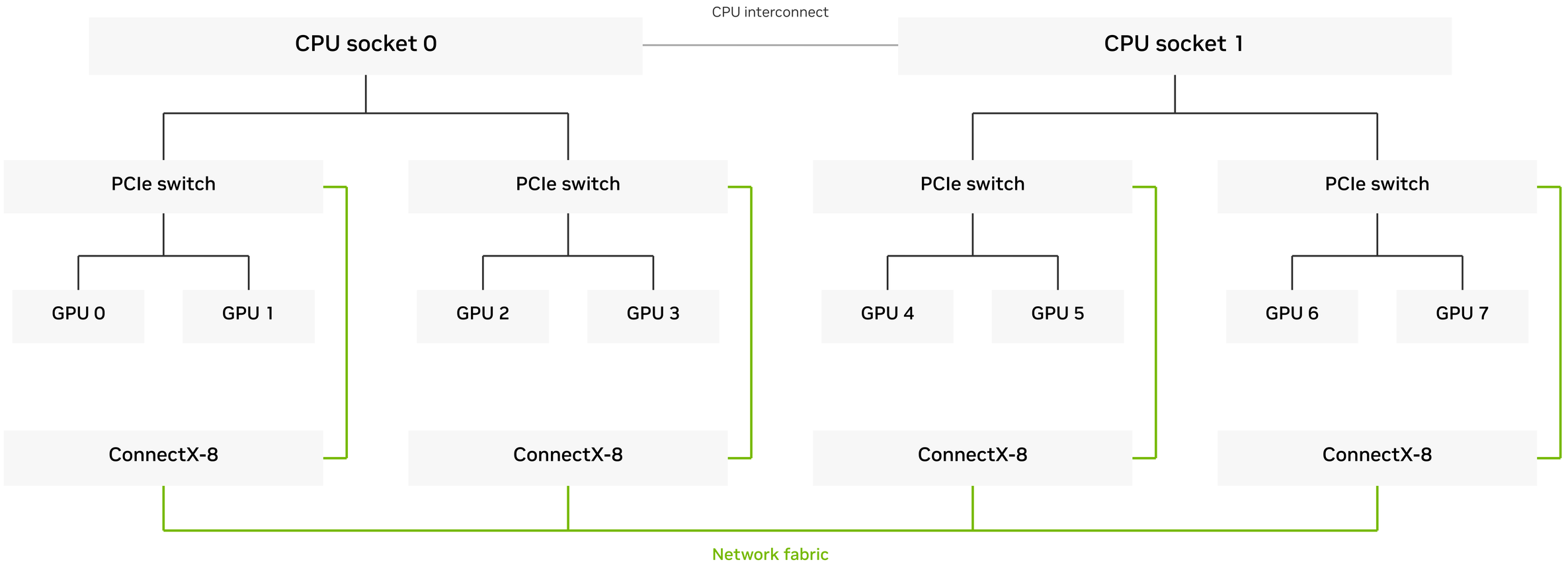
Train for quantization error.
Keep sensitive paths at higher precision.

Evaluate speed and quality together on representative scenes.

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Optimization - Comms

Map GPU–NIC affinity before tuning the collective.



8 × PRO 5000 · 4 × ConnectX-8 · two GPUs per PCIe subtree · no GPU NVLink

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Optimization - Comms

Choose the path, overlap the work, and reduce movement.

Route

Use GPU-affine NICs and RDMA.

Overlap

Transfer the next head band while attention runs.

Streamline

Use direct layouts and replay stable compute.

Overlap transfers with attention



Conceptual head-band schedule

3.1× faster

Grouped E2E: 19.46 s → 6.19 s

Route efficiently. Overlap work. Reduce movement.

Grouped result: RDMA + layout + head overlap + compute graphs.

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From Minutes to Real Time

960 × 544 · 5.167 s output · 8 × RTX PRO 5000

43.4× faster

213.40 s → 4.92 s

Recorded baseline · 50 evaluations

Optimized H3 · 8 evaluations

5.17 s of video generated in 4.92 s · RTF 0.95

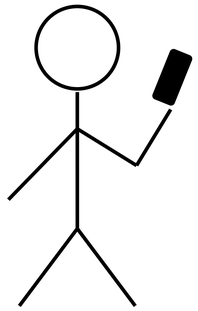
Recorded E2E medians; baseline measured separately. Includes model, schedule and precision changes. Standalone H3.

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From a Question to a Reply

Voice → text → prompt → audio + video

"Hi, could you introduce yourself?"



User

Voice

Qwen3-ASR

0.6B · Speech → text

Qwen3-4B

Reply → H3 prompt

Reference image



H3

Audio + video

"Hi, I'm Miles..."



Avatar reply

PROMPT EXCERPT

... "Hi, I'm Miles. I help with simple, clear answers." ... His lips follow the words.

Hear the avatar's reply

960 × 544 · ~5 s windows · 8 steps · 8 × RTX PRO 5000

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Live Demo

A continuous avatar conversation.



960 × 544 · ~5 s windows · 8 steps · 8 × RTX PRO 5000

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Takeaways

01

Optimize the communication path.

Topology, bandwidth and less data movement.

02

Validate quantization case by case.

Measure quality and latency together.

03

Next: responsive, richer interaction.

Autoregressive diffusion + prompt enhancement.

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